nRF52832-Based Robotic Kit **Datasheet**

The nRF52832-Based Robotic Kit is a high-tech development platform designed for students, hobbyists, and researchers to explore advanced robotics and IoT applications. Powered by the nRF52832 MCU (ARM Cortex-M4F) with Bluetooth 5.0, it integrates an L293D motor driver, OLED display, and sensor interfaces. USB Type-C programming, onboard power regulation, and modular connectivity make it perfect for hands-on learning, prototyping, and smart robotic projects.

Key Features

- High-Performance MCU: Powerful ARM Cortex-M4F processor for smooth and fast robot control
- Advanced Bluetooth 5.0: Reliable wireless connectivity for real-time robot operation and IoT integration
- Versatile Motor Control: Supports multiple motors for building dynamic robotic systems
- Live Data Display: Onboard OLED interface for real-time feedback and visualization
- Expandable Sensor Support: Easy connection for gas, ultrasonic, and analog sensors
- Flexible Power Options: USB-C and external power support with built-in protection
- Modular I/O & PWM: Expandable pins for connecting additional sensors, servos, and peripherals
- Interactive Feedback: Buzzer and LEDs for alerts and system status
- Educational & Prototype Ready: Ideal for learning, experimentation, and advanced student robotics projects

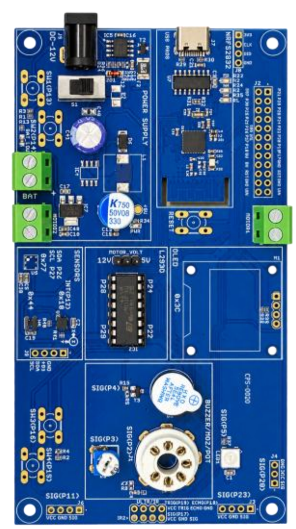

Fig: Robotic Kit

Technical Specifications

Parameter	Value
Microcontroller	Nordic nRF52832 (ARM Cortex-M4F, 64 MHz, 512 KB Flash, 64 KB SRAM)
Wireless Connectivity	Integrated BLE 5.0 for fast, reliable Bluetooth communication
Motor Driver	L293D Dual H-Bridge, 2 DC motors or 1 stepper, 4.5–36 V, 600 mA/channel, 1.2 A peak
Display	0.96″ OLED (I²C) for live data and sensor feedback
Sensors	MQ gas sensor header, ultrasonic sensor interface, potentiometer for analog calibration
Power	USB Type-C programming/power, 7–12 V DC input, onboard 3.3 V/5 V regulators, reverse-polarity protection
Expansion	Multiple GPIO, PWM, and ADC pins, motor terminal blocks for extra motors
Extras	Onboard buzzer, reset button, and power/user LEDs for real-time feedback

Applications

- Smart Robotics Projects – build line followers, obstacle-avoiding, and Bluetooth-controlled robots
- IoT & Mobile Robots – integrate sensors and wireless control for smart projects
- STEM Learning Platform – hands-on education for students in embedded systems, programming, and electronics
- Sensor-Based Experiments – gas detection, distance sensing, and environmental monitoring
- Prototyping & Innovation – rapid development of smart automation systems and robotic prototypes
- Educational Competitions – perfect for robotics contests, STEM workshops, and lab exercises